

# Answer Key — Biology Grade 11

Sub-Strand 1.4: Cell Division • Lesson 6: Budding and Binary Fission: Making New Cells and Organisms from One Parent

Teacher reference only. Do not project or hand out.

## Quick marking grid

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 |
|----|----|----|----|----|----|----|
| B  | B  | A  | A  | A  | D  | C  |

USE AFTER: PREDICT PHASE — Predict: think-pair-share on how a single yeast cell or bacterium produces a new organism without sex cells

**Q1. A single yeast cell in dough can become millions of cells in a few hours. Which statement best explains how one yeast cell can produce a new organism?**

- A. A male and a female yeast cell must be present
- B. The new organism arises from one parent cell without fusion of sex cells**
- C. Two yeast cells must first fuse their sex cells
- D. The yeast cell must be fertilised by food such as sugar

**Correct answer: B.** Yeast reproduces asexually, so a single parent makes a new organism without sex cells fusing. The other options wrongly assume two parents or fertilisation.

USE AFTER: OBSERVE PHASE — Pair B (Bacteria): plasticine and string model of binary fission

**Q2. A rod-shaped bacterium reproduces by binary fission. Which sequence correctly shows what happens?**

- A. The cell splits in two, and only one daughter cell receives the chromosome
- B. The chromosome is copied, the cell lengthens, a dividing wall forms, and two equal cells result**
- C. The bacterium fuses with another bacterium to form a new cell
- D. A small bud grows on the cell, receives a copy of the nucleus and then breaks off

**Correct answer: B.** In binary fission the circular chromosome is copied before the cell elongates, forms a dividing wall and splits into two equal daughter cells. The bud option describes budding in yeast.

USE AFTER: OBSERVE PHASE — Doubling question: if one bacterium divides every 20 minutes, how many after 2 hours?

**Q3. One bacterium in contaminated water divides into two every 20 minutes. Assuming every cell keeps dividing, how many bacteria will there be after 2 hours?**

- A. 64**
- B. 32
- C. 6
- D. 12

**Correct answer: A.** 2 hours = 120 minutes, so there are  $120 \div 20 = 6$  divisions. The number doubles each time: 1, 2, 4, 8, 16, 32, 64, which is  $2^6 = 64$ .

USE AFTER: EXPLAIN PHASE — Discussion diamond: ranking the nine statement cards to answer how a single organism can produce a new organism without sex cells

**Q4. Which statement is true for BOTH budding in yeast and binary fission in bacteria?**

- A. Both copy the genetic material before the cell divides, and the new organism comes from one parent**
- B. Both produce offspring that are genetically different from the parent
- C. Both always produce two daughter cells of equal size
- D. Both need the fusion of sex cells from two parents

**Correct answer: A.** Both are asexual reproduction: genetic material is copied first, one parent is involved, and the daughter cells are genetically identical to the parent. Only binary fission gives equal cells, and neither process uses sex cells.

USE AFTER: EXPLAIN PHASE — Discussion diamond teacher question: How is this like what happens when a baby grows? How is it different?

**Q5. A child grows from 3 kg to 50 kg by cell division. How is this similar to, but different from, a bacterium dividing by binary fission?**

- A. In both cases the cells divide, but in the child the new cells add to one body, while in a bacterium each new cell becomes a new organism**
- B. Only the child's cells copy their genetic material before dividing
- C. Only the bacterium's division needs two parents
- D. Both add cells to one body, so the cells stay joined together

**Correct answer: A.** Both involve cells copying their instructions and dividing. In a growing child the new cells join the same body, whereas in a single-celled organism the new cells are separate new organisms.

USE AFTER: DQB PHASE — Gallery walk and exit ticket: write one difference between budding and binary fission and one similarity

**Q6. Which is a correct difference between budding in yeast and binary fission in bacteria?**

- A. Budding involves sex cells, while binary fission does not
- B. Budding happens in bacteria, while binary fission happens in yeast
- C. Budding needs two parents, while binary fission needs one parent
- D. Budding gives a smaller bud and a larger parent, while binary fission gives two equal cells**

**Correct answer: D.** In budding a small bud grows on the parent yeast cell and separates, so the cells are unequal. In binary fission the bacterium splits into two equal cells. Both use one parent and no sex cells.

USE AFTER: MODEL BUILDING PHASE — Revised model panel: health link between fast bacterial multiplication, boiling water and handwashing

**Q7. Bacteria that cause cholera and typhoid can multiply very quickly in unclean water. Why does boiling drinking water help prevent these diseases?**

- A. Boiling turns the bacteria into yeast cells that are harmless
- B. Boiling allows the bacteria to fuse and become weaker
- C. Boiling kills the bacteria, so they cannot divide and increase in number**
- D. Boiling makes the bacteria stop copying their DNA but they stay alive

**Correct answer: C.** Bacteria increase in number by binary fission, which doubles their numbers quickly. Boiling kills them, so they cannot divide and cause disease.